Improving Human-Exoskeleton Coordination through Modification of the Torque's Rise and Fall Time Control Parameters in Powered Ankle Exoskeleton

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Abstract

Exoskeletons have become increasingly popular due to their potential to enhance or assist human capabilities across various industries, such as manufacturing and healthcare. Lower extremity-powered exoskeletons are driven by controllers that store commands to assist the user. However, they still have their own unique set of complexities and unknowns. It needs to be determined how changes in exoskeleton control parameters can affect human-exoskeleton coordination. Changes in exoskeleton parameters, such as ankle timing and quasi-stiffness, can influence gait characteristics and muscle activation, affecting human performance. This research aims to develop a new exoskeleton controller for the Dephy ExoBoot to modify the torque profile's rise and fall time (exoskeleton parameters) and evaluate people's sensitivity to the changes and associated changes in gait characteristics. This controller will be used in a pilot study to provide a preliminary understanding of human perception towards changes in exoskeleton control parameters and insight into individual differences in exoskeleton usage. The preliminary study results will inform exoskeleton precision requirements to maximize human-system interaction, improving human performance.

Introduction

As powered exoskeletons become more advanced and widely adopted, the ability to tailor assistance to individual users has become a key challenge in improving their effectiveness. While most current exoskeleton systems rely on fixed torque parameters, human gait is inherently variable, and assistance that lacks adaptability may reduce performance or even hinder the user experience.

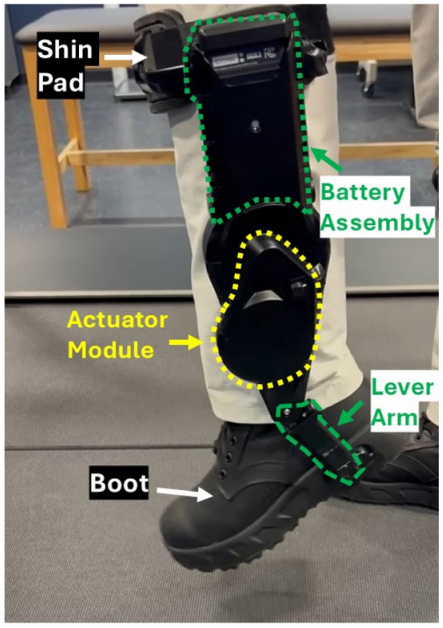

Fig. 1 Dephy ExoBoot (Dephy Inc, Maynard, MA)

In powered ankle exoskeletons, the torque profile timing plays a critical role in how natural and helpful the user perceives the assistance. However, existing controllers do not allow these parameters to be easily modified or personalized. It is also not clear how precise these controllers will need to be to assist the user. Therefore, a new torque controller for the Dephy ExoBoot was developed to allow for programmable preset values of the rise and fall times within a fixed actuation window. This enables future testing to evaluate individual responses to these changes both biomechanically and perceptually. By enabling control of torque dynamics, this controller is a platform for advancing user-specific exoskeleton

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strategies that promote comfort, coordination, and biomechanical efficiency.

The Dephy ExoBoot (Figure 1) provides torque/assistance at the push-off during the stance phase of the gait cycle. To provide the assistance, the actuator produces a torque about the ankle joint by spooling a rigid chain attached to the exoskeleton's lever arm.

Current Controller

The Dephy Exoskeleton ankle torque profile (Figure 2) consists of three key parameters: actuation start, peak torque and actuation end [1, 2]. These three parameters are used for a cubic spline fitment of the torque profile slope. The peak torque has two parameters that can be modified: rise time and fall time.

The ankle torque profile begins at the user's detection of a heel strike via ExoBoot's onboard IMU. Time is defined as percentage of gait stride period. The ankle torque profile requires two controllers and is divided into four regions. Regions 1 & 4 are under position control to keep the chain under tension without applying torque. Regions 2 & 3 are under current control for torque actuation. Actuation start is the transition point between regions 1 & 2 where torque begins to be applied. Peak torque is the transition between the rise and fall of the torque slopes, respective to user weight and the ExoBoot's given mechanical constraint. Actuation end is the transition point between regions 3 & 4 where the torque is back to 0.

Current control is shown in Equation 1:

$$I_m = \frac{T_m}{k_t} \tag{1}$$

where T_m is the motor torque, I_m is the motor current & k_t is defined as 0.14 Nm/A.

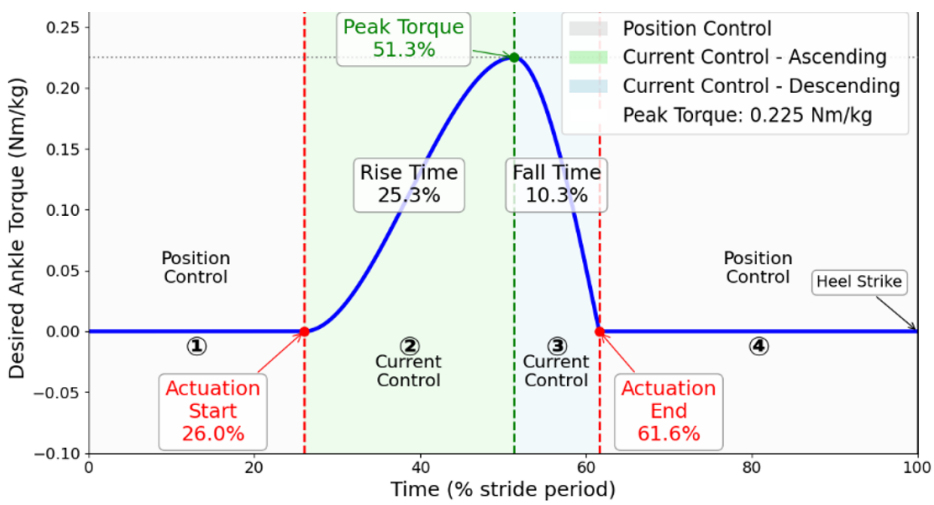

Fig. 2 Exoskeleton Torque Profile

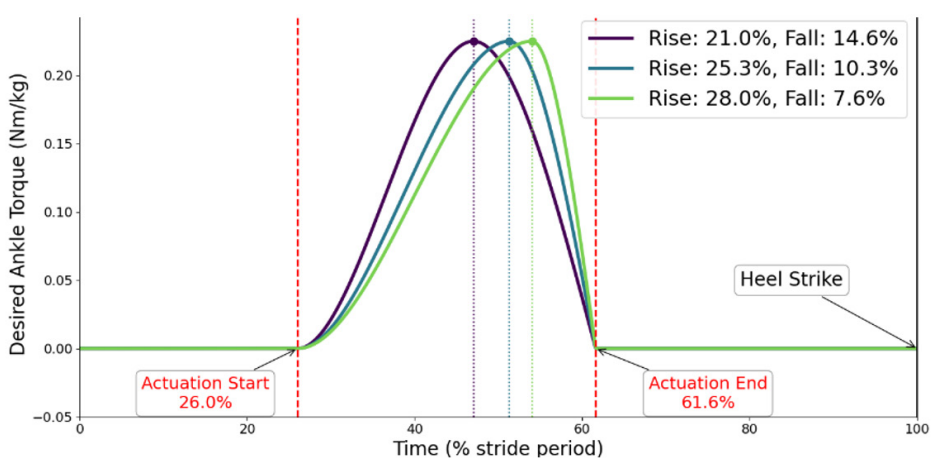

Fig. 3 Comparison of different Rise/Fall time parameters

New Controller

A new controller was developed allowing for adjustable rise and fall times, a largely unexplored area in exoskeleton research. The actuation start and end times remained constant in the controller with values 26% stride period and 61.6% stride period allowing for gait actuation to remain the same. Peak torque is also kept constant at 0.225 Nm/Kg. Allowing for only the rise time and/or fall time parameters to be able to be adjusted and isolated as demonstrated in Figure 3. Meaning that the entire adjustable time for this controller is 35.6%. A weight of 72.3 Kg was used as a reference weight based on data collected in previous research [1,2].

This new controller was developed in Python using the Dephy Inc. FlexSEA Python API and tested on the Dephy ActPack actuator module using a Raspberry Pi (Model: 4B). Eventually it be integrated into the Dephy ExoBoot to conduct the preliminary study.

Our hypothesis for the development of this controller and preliminary study is that if the torque profile's rise and fall time does not align with an individual's toe off during the gait cycle, the exoskeleton assistance may be perceived as uncomfortable or inefficient. For example, if the torque rises too quickly, it can result in more abrupt assistance that feels unnatural. On the other hand, if the torque falls too slowly, it may provide prolonged support that introduces unwanted resistance. These mismatches can negatively impact user experience and coordination. This new controller lays the groundwork for further understanding of how changes in the torque curve influence an individual's perception & gait dynamics which is crucial for optimizing performance, enhancing comfort, and improving overall human-exoskeleton interaction.

Future Direction

This controller advances exoskeleton technology by enabling and exploring the changes in user behaviors due to adjustments to the torque profile's rise and fall times—an area that has been largely unexplored in previous research. The controller will be used in an upcoming study involving human trials, where participants will walk with the ExoBoot under varied torque transition conditions. These trials aim to evaluate individuals' sensitivity to changes in the torque profile and analyze associated changes in gait characteristics, comfort, and overall coordination. The findings from this study will provide valuable insight into how users perceive different torque dynamics and will help guide future exoskeleton control strategies aimed at improving user trust, comfort, and efficiency.

Statement of Research Advisor

Max has demonstrated outstanding proficiency in coding the exoskeleton. He not only exhibited his coding abilities but also his problem-solving skills, creativity, and attention to detail while developing the code. The undergrad research conducted by Max has excellent potential for contributing to the robotics and biomechanics field.

-Yadrianna Acosta-Sojo, Department of Industrial and Systems Engineering, Samuel Ginn College of Engineering

References

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[2] Peng, Xiangyu, et al. "Perception of Powered Ankle Exoskeleton Actuation Timing During Walking: A Pilot Study." 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), IEEE, 2021, pp. 4654–57. DOI.org (Crossref), https://doi.org/10.1109/EMBC46164.2021.9629925.

Authors Biography

Max is a senior undergraduate student in Industrial and Systems Engineering at Auburn University. He is an undergraduate research assistant at the Center for Occupational Safety, Ergonomics, and Injury Prevention in the Industrial and Systems Engineering Department. Max's research focuses on advanced control algorithms and integrating new controllers onto powered ankle exoskeletons for gait assistance. He also holds an Undergraduate Research Fellowship under the Office of Undergraduate Research. He is involved in various clubs and organizations on campus. He has previously done a Co-op at GE Aerospace, where he developed AI-based inspection tools and process automation solutions.

Yadrianna Acosta-Sojo is an Assistant Professor of Industrial and Systems Engineering at Auburn University. She is also a faculty member of the Center for Occupational Safety, Ergonomics, and Injury Prevention (COSEIP) and an affiliated faculty member of the Interdisciplinary Center for Advanced Manufacturing Systems (ICAMS). Her expertise includes ergonomics, human factors, wearable sensors, experimental biomechanics, and assistive technologies. Acosta's research objectives include utilizing wearable technologies and methods to (1) understand changes in proprioception and motor control, (2) improve rehabilitation for older adults and patients with neurological disorders, and (3) reduce the risk of injuries or mitigate age-related changes in sensorimotor control to ensure people's safety, health, and well-being.